Experimental reproducibility limits mRNA-protein correlation in tumor profiles

BBCC2021 – online edition
http://www.bbcc-meetings.it

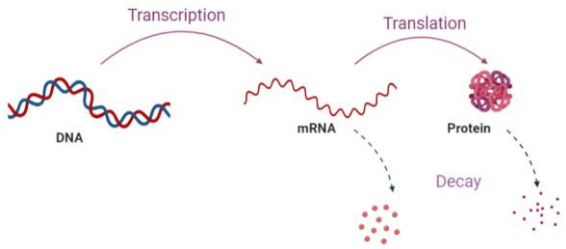
Swathi Ramachandra Upadhya[1], Colm J Ryan[1]

[1]School of Computer Science and Systems Biology Ireland, University College Dublin, Ireland

Read more in our preprint on bioRxiv:
https://doi.org/10.1101/2021.09.22.461108

INTRODUCTION

- Proteins – workhorses of cellular functions with varying abundances across tumor cells.
- Quantifying proteins – useful for precision medicine, but it is technically challenging.
- mRNAs, precursor of proteins, are used as proxies for protein measurements.



Central dogma of molecular biology

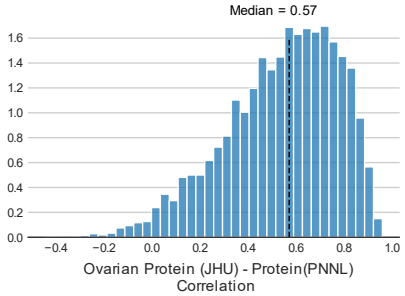
- mRNA-protein correlation identified by multiple studies to be only moderate (~0.4).
- The unexplained variation between mRNA-protein correlation is attributed to post-transcriptional regulation and measurements errors/noises.

RESEARCH QUESTION

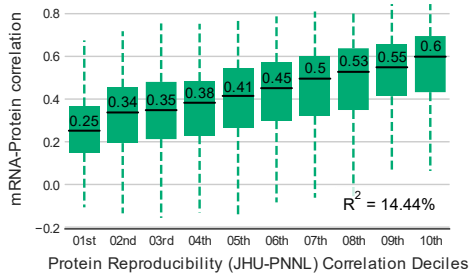
Can we determine the extent to which the unexplained variation in mRNA-to-protein correlation is due to measurement errors / noise?

RESULTS

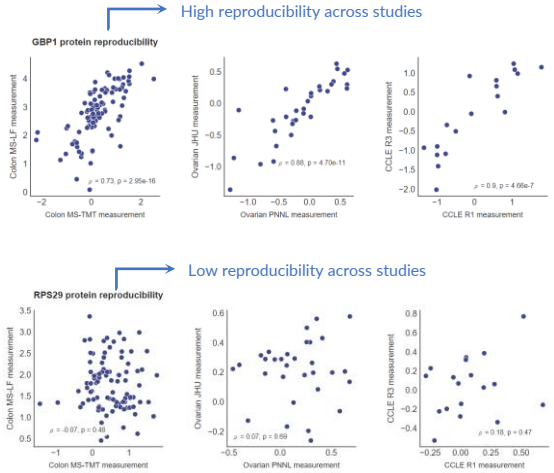
1. Protein reproducibility correlation (calculated by seeing how reproducible a given protein is across replicates) varies significantly.



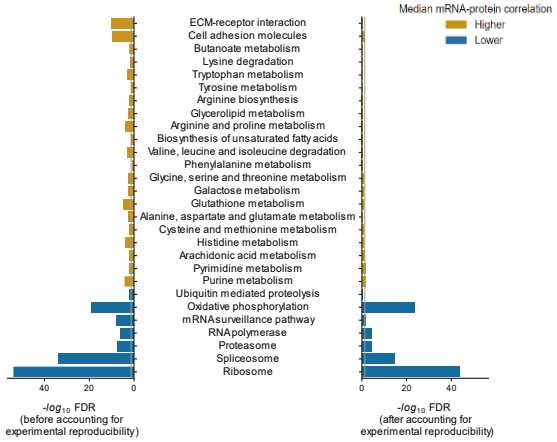
2. mRNA-to-protein correlation increases with protein reproducibility correlation.



3. Protein reproducibility is consistent across studies



4. Metabolic pathways associated with higher-than-average mRNA-protein correlation reflect differential experimental reproducibility



METHODS

We focused on proteomics as large-scale quantification of proteins has lagged behind the large-scale sequencing of genomes and transcriptomes. Using 3 studies (ovarian, colon tumor studies and CCLE) with replicate proteomic profiles, we assessed the variation in mRNA-protein correlation with respect to protein reproducibility. Validated result (2) in ten other studies without replicate proteomic profiles by aggregating protein reproducibility. Finally, we performed pathway enrichment analysis accounting for experimental reproducibility.

CONCLUSION

- Significant variation in the reproducibility of protein measurements across samples. Some proteins tend to have higher reproducibility than other proteins.
- A major source of variation in mRNA-protein correlation is experimental reproducibility.
- Pathways associated with higher-than-average mRNA-protein correlation reflect differential experimental reproducibility rather than differential post-transcriptional regulation.

REFERENCES

Zhang, Hui, et al. "Integrated proteogenomic characterization of human high-grade serous ovarian cancer." Cell 166.3 (2016): 755-765.

Vogel, C., & Marcotte, E. M. (2012). Insights into the regulation of protein abundance from proteomic and transcriptomic analyses. *Nature reviews genetics*, *13*(4), 227-232.